
Lab 2: Adversarial Example Generation

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Project Repository:

github.com/luisy-eng/ECE491_Lab2

Abstract

This project investigates adversarial attacks against a machine learning classifier using the Fast Gradient Sign Method (FGSM). First, an untargeted FGSM attack is applied to a pretrained model trained on the MNIST handwritten digit dataset. The effect of different perturbation strengths is evaluated by measuring classification accuracy and visualizing adversarial examples. Next, FGSM is modified to perform a targeted attack in which adversarial examples are generated with the goal of forcing the classifier to predict class 5. Finally, several defense techniques, including denoising and noise-based preprocessing, are evaluated to determine their ability to reduce the effectiveness of the adversarial attacks.

1 Introduction

Machine learning models can achieve high classification accuracy while remaining vulnerable to small changes in their input data. Adversarial examples are inputs that have been intentionally modified to cause a machine learning model to produce an incorrect prediction [2].

In this project, adversarial examples are generated using the Fast Gradient Sign Method (FGSM), a gradient-based adversarial attack introduced by Goodfellow et al. [2].

The project is divided into three tasks. First, the standard untargeted FGSM attack is demonstrated using the MNIST handwritten digit dataset. Second, the attack is modified into a targeted FGSM attack designed to cause the model to classify adversarial examples as the digit 5. Third, different preprocessing defenses are evaluated to determine whether they can reduce the effectiveness of the attack.

2 Task 1: FGSM Demonstration

2.1 Fast Gradient Sign Method

The FGSM demonstration was implemented in Python using PyTorch and Google Colab. The experiment used the MNIST handwritten digit dataset [3] and followed the PyTorch adversarial example generation tutorial [4].

$$x_{\text{adv}} = x + \epsilon \text{sign}(\nabla_x J(\theta, x, y)), \quad (1)$$

where x represents the original input image, y is the true class label, θ represents the parameters of the neural network, and J is the loss function.

The gradient

$$\nabla_x J(\theta, x, y) \quad (2)$$

describes how the loss changes with respect to the pixels of the input image. FGSM takes the sign of this gradient and scales it using the perturbation parameter ϵ .

The parameter ϵ determines the strength of the adversarial perturbation. Larger values of ϵ modify the input image more strongly and generally result in a higher probability of misclassification.

2.2 Experimental Setup

The FGSM demonstration was implemented in Python using PyTorch and Google Colab. The experiment used the MNIST handwritten digit dataset and a pretrained convolutional neural network classifier.

The MNIST test dataset was used to evaluate the network. Only samples that were originally classified correctly were used to generate adversarial examples. For each image, the gradient of the loss with respect to the input pixels was calculated through backpropagation.

The attack was evaluated using the following perturbation strengths:

$$\epsilon = \{0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}. \quad (3)$$

The adversarial image was generated by adding the signed gradient to the original image and scaling it according to ϵ . The resulting pixel values were clipped to the valid image range.

2.3 Results

The FGSM experiment demonstrated that the performance of the classifier decreased as the magnitude of the adversarial perturbation increased. When $\epsilon = 0$, the input images were unchanged and the model maintained its normal classification accuracy.

As ϵ increased, more input images were incorrectly classified. This shows that relatively small changes to an input image can significantly affect the output of a neural network.

Figure 1 shows the relationship between ϵ and classification accuracy.

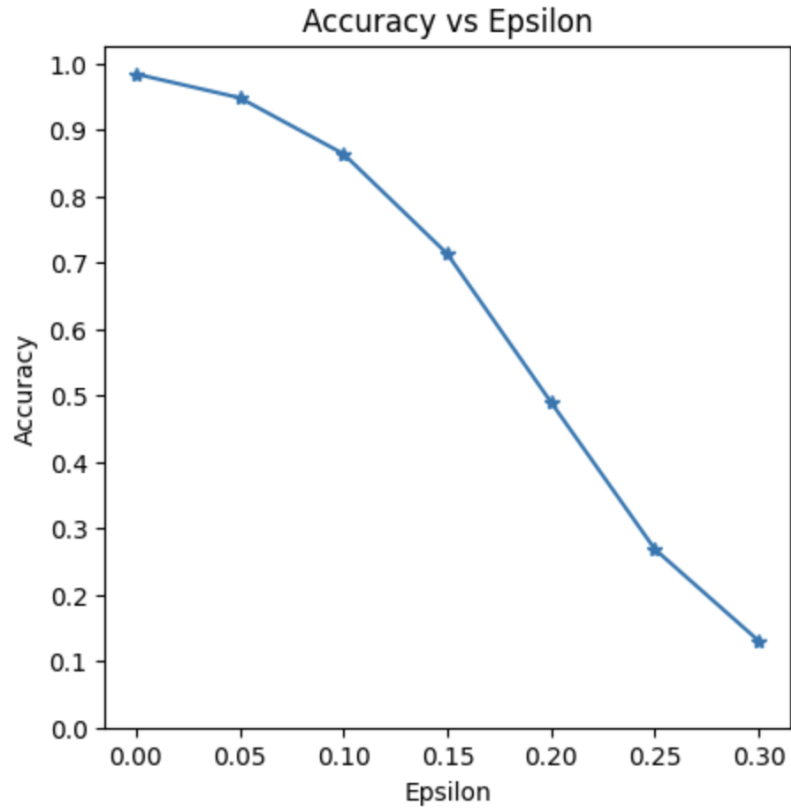


Figure 1: Classification accuracy of the MNIST model for different FGSM perturbation strengths.

Examples of generated adversarial images are shown in Figure 2. As ϵ increases, the perturbation becomes increasingly visible. However, several images remain recognizable to a human observer even when the neural network produces an incorrect classification.



Figure 2: MNIST adversarial examples generated using FGSM for different values of ϵ .

These results demonstrate that the classifier is vulnerable to gradient-based adversarial perturbations.

3 Task 2: Targeted FGSM Attack

3.1 Objective

The FGSM attack implemented in Task 1 is an untargeted adversarial attack. In an untargeted attack, the objective is simply to cause the classifier to produce a prediction that is different from the correct class. The specific incorrect class does not matter.

In Task 2, the FGSM attack was modified into a targeted adversarial attack. Instead of only attempting to cause a misclassification, the objective was to force adversarial examples toward one specific target class. For this experiment, the target class was selected as the digit 5,

$$y_{\text{target}} = 5. \quad (4)$$

Therefore, images belonging to the MNIST classes

$$\{0, 1, 2, 3, 4, 6, 7, 8, 9\} \quad (5)$$

were modified with the objective of causing the neural network to classify them as the digit 5.

Images whose true class was already 5 were excluded from the targeted attack evaluation because these samples already belonged to the desired target class and therefore would not represent a successful adversarial misclassification.

3.2 Targeted FGSM Method

The standard untargeted FGSM attack generates an adversarial example according to

$$x_{\text{adv}} = x + \epsilon \text{sign}(\nabla_x J(\theta, x, y)), \quad (6)$$

where x is the original input image, y is the correct class, θ represents the model parameters, J is the loss function, and ϵ controls the magnitude of the perturbation.

In the untargeted case, the perturbation is added in the direction that increases the loss associated with the correct class. This encourages the model to move away from the original prediction.

For the targeted attack, the objective is different. Instead of increasing the loss associated with the correct class, the goal is to decrease the loss associated with a selected target class. Therefore, the direction of the FGSM perturbation is reversed:

$$x_{\text{adv}} = x - \epsilon \text{sign}(\nabla_x J(\theta, x, y_{\text{target}})). \quad (7)$$

For this experiment,

$$y_{\text{target}} = 5. \quad (8)$$

The negative sign in Equation 7 moves the input image in the direction that reduces the classification loss for the target class. As a result, the adversarial perturbation attempts to move the model prediction toward class 5 rather than simply away from the original class.

3.3 Implementation

The targeted attack was implemented by modifying the FGSM procedure used in Task 1. The same pretrained MNIST classifier and test dataset were used so that the effect of the targeted attack could be compared directly with the original FGSM experiment.

For each test image, the model first produced an initial prediction. Only samples that were classified correctly before the attack were used for the targeted attack experiment. This ensured that any later misclassification was caused by the adversarial perturbation rather than an existing model error.

A new target tensor containing the label 5 was then created. Instead of calculating the loss using the true class label, the loss was calculated with respect to the target class:

$$J_{\text{target}} = J(\theta, x, 5). \quad (9)$$

Backpropagation was then used to calculate the gradient of the target loss with respect to the input image,

$$\nabla_x J_{\text{target}}. \quad (10)$$

Because the MNIST images were normalized before being passed into the network, the images were converted back to their original pixel scale before the adversarial perturbation was applied. The targeted FGSM perturbation was then generated using Equation 7.

After the perturbation was added, each pixel was clipped to the valid MNIST image range,

$$0 \leq x_{\text{adv}} \leq 1. \quad (11)$$

The adversarial image was then normalized again before being passed through the neural network for classification.

The attack was evaluated using the following perturbation strengths:

$$\epsilon = \{0.00, 0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}. \quad (12)$$

The targeted attack was considered successful only when the final model prediction was exactly equal to the selected target class:

$$\hat{y}_{\text{adv}} = 5. \quad (13)$$

The targeted attack success rate (ASR) was calculated as

$$\text{ASR} = \frac{N_{\text{classified as 5}}}{N_{\text{eligible attack samples}}}, \quad (14)$$

where $N_{\text{classified as 5}}$ represents the number of successfully targeted adversarial samples and $N_{\text{eligible attack samples}}$ represents the number of correctly classified test samples whose original class was not 5.

3.4 Results

A total of 8965 correctly classified MNIST test images were eligible for the targeted attack. Samples belonging to the original class 5 were excluded from the evaluation.

Table 1 summarizes the measured targeted attack success rate for each tested value of ϵ .

Table 1: Targeted FGSM attack results for target class 5.

ϵ	Successful Attacks	Success Rate (%)
0.00	0 / 8965	0.00
0.05	40 / 8965	0.45
0.10	184 / 8965	2.05
0.15	535 / 8965	5.97
0.20	1258 / 8965	14.03
0.25	2355 / 8965	26.27
0.30	3510 / 8965	39.15

At $\epsilon = 0$, no perturbation was applied to the input images. Consequently, none of the eligible images were changed to the target class, resulting in a targeted attack success rate of 0%. This result also served as a sanity check for the implementation because the model prediction should not change when the perturbation magnitude is zero.

At $\epsilon = 0.05$, only 40 of the 8965 attacked samples were successfully classified as the digit 5, producing a success rate of 0.45%. Increasing the perturbation strength to $\epsilon = 0.10$ increased the number of successful attacks to 184, corresponding to a success rate of 2.05%.

The attack became considerably more effective for larger values of ϵ . At $\epsilon = 0.20$, 1258 samples were successfully forced into the target class, resulting in a success rate of 14.03%. Increasing ϵ to 0.25 produced a success rate of 26.27%.

The highest targeted attack success rate occurred at $\epsilon = 0.30$. At this perturbation strength, 3510 of the 8965 eligible samples were classified as the target digit 5, resulting in an attack success rate of 39.15%.

Figure 3 shows the relationship between the perturbation magnitude and the targeted attack success rate.

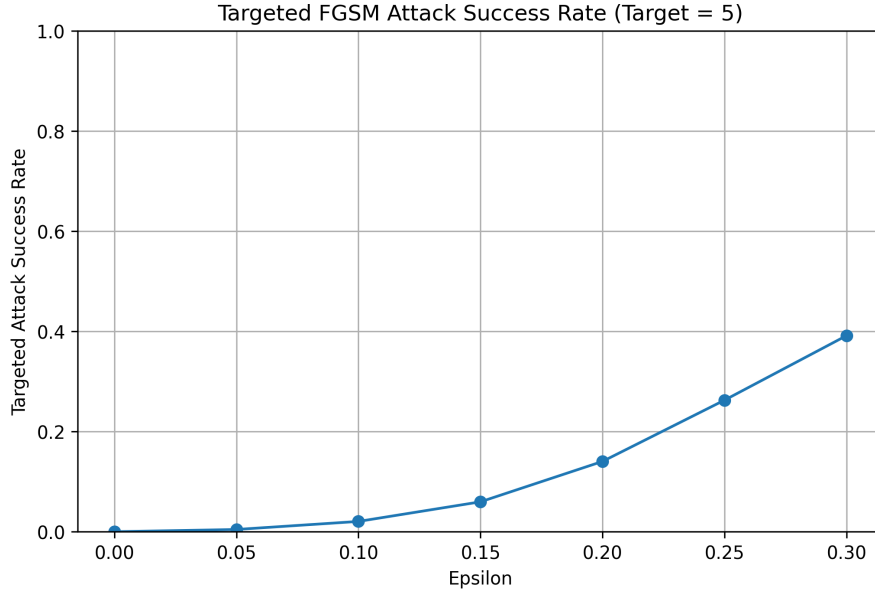


Figure 3: Targeted FGSM attack success rate for target class 5 as the perturbation strength ϵ increases.

The results show a clear relationship between perturbation strength and targeted attack effectiveness. Small perturbations produced relatively few successful targeted attacks, while larger perturbations greatly increased the probability that an input would be classified as the digit 5.

The increase was also nonlinear. The attack success rate increased from 5.97% at $\epsilon = 0.15$ to 14.03% at $\epsilon = 0.20$, and then to 39.15% at $\epsilon = 0.30$. This indicates that larger perturbations allow targeted FGSM to move a greater number of inputs across the decision boundary associated with the target class.

Figure 4 shows adversarial examples generated for different values of ϵ . Each column represents a different original MNIST digit, while each row represents a different perturbation strength.

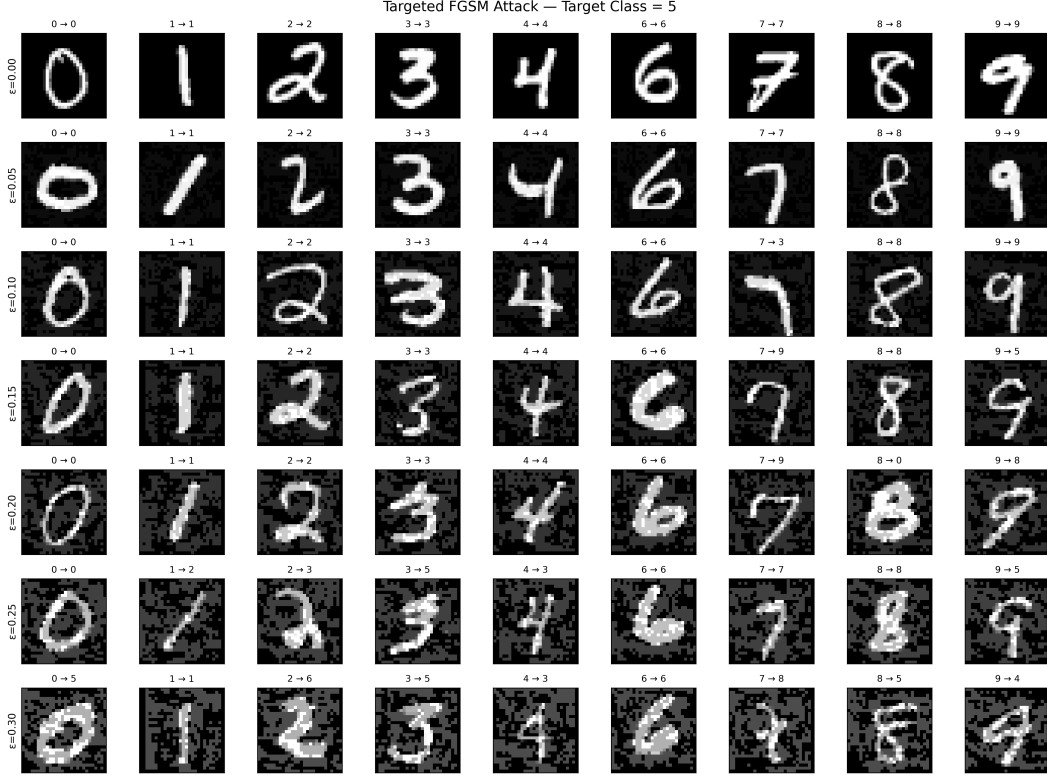


Figure 4: Examples of targeted FGSM adversarial images generated for perturbation strengths from $\epsilon = 0$ to $\epsilon = 0.30$. The attack objective is to move the model prediction toward target class 5.

For small perturbation values, most samples remained correctly classified. As ϵ increased, the effect of the adversarial perturbation became stronger and additional samples changed class.

The visual results also illustrate an important tradeoff in adversarial example generation. Increasing ϵ improves the probability of a successful attack, but the perturbation also becomes increasingly visible in the image. At larger values of ϵ , the structure of some handwritten digits becomes noticeably distorted.

To verify that the targeted attack was capable of producing successful examples from the different source classes, successful adversarial samples were also collected at $\epsilon = 0.30$.

Figure 5 shows examples in which the original digit was successfully moved to target class 5.

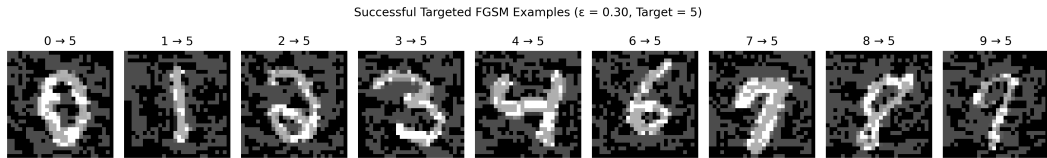


Figure 5: Successful targeted FGSM examples at $\epsilon = 0.30$. Each adversarial image is classified by the neural network as the target digit 5.

3.5 Discussion

The results demonstrate that FGSM can be modified successfully from an untargeted attack into a targeted attack by changing the loss label to the desired target class and reversing the direction of the gradient perturbation.

The attack success rate increased consistently as the allowable perturbation magnitude increased. At low values of ϵ , the perturbation was generally insufficient to move the model prediction all the way from the original class to the selected target class. However, as ϵ increased, the input was moved farther in the direction that reduced the loss associated with class 5.

Even at the largest tested value of $\epsilon = 0.30$, the targeted attack success rate reached 39.15% rather than 100%. This highlights an important difference between targeted and untargeted adversarial attacks. An untargeted attack only needs to move an input away from its correct classification, while a targeted attack must move the input toward one particular class. The targeted attack therefore places a more restrictive requirement on the adversarial perturbation.

Overall, the experiment demonstrates that relatively small, gradient-based modifications to an input can intentionally influence the output of a neural network toward a selected target class. The results also demonstrate the tradeoff between attack effectiveness and perturbation magnitude: increasing ϵ improves the probability of a successful targeted attack but also produces more visible changes to the input image.

4 Task 3: Adversarial Defenses

4.1 Objective

The objective of Task 3 was to investigate whether simple input preprocessing techniques could reduce the effectiveness of the targeted FGSM attack developed in Task 2.

Two categories of defenses were evaluated. The first category consisted of denoising techniques designed to remove or smooth small adversarial perturbations. Common image-processing operations such as Gaussian, mean, and median filtering are widely used for smoothing and noise reduction [1]. Three denoising methods were tested:

- Gaussian blur,
- mean filtering,
- median filtering.

The second category consisted of adding random Gaussian noise to the adversarial image. Several noise levels were evaluated to determine how the amount of added noise affected the targeted attack success rate.

All defenses were evaluated against the targeted FGSM attack with

$$\epsilon = 0.30 \tag{15}$$

and target class

$$y_{\text{target}} = 5. \tag{16}$$

The same pretrained MNIST classifier and the same set of eligible test samples used in Task 2 were used in this experiment.

4.2 Experimental Procedure

For each correctly classified MNIST test image whose true class was not 5, a targeted FGSM adversarial example was first generated using the procedure from Task 2.

The attack was generated according to

$$x_{\text{adv}} = x - \epsilon \text{sign}(\nabla_x J(\theta, x, y_{\text{target}})). \tag{17}$$

The resulting adversarial image was then passed through one of the defense methods before being reclassified by the neural network.

The attack was considered successful after defense if the final model prediction remained equal to the target class 5.

Therefore, the attack success rate after defense was calculated as

$$\text{ASR}_{\text{defense}} = \frac{N_{\text{predicted as 5}}}{N_{\text{eligible samples}}}. \quad (18)$$

A lower attack success rate indicates a more effective defense.

4.3 Denoising Defenses

Three different denoising methods were applied to the adversarial images.

4.3.1 Gaussian Blur

Gaussian blur smooths the image using a Gaussian kernel. The purpose of this preprocessing step is to reduce high-frequency pixel variations that may have been introduced by the adversarial perturbation.

A 3×3 Gaussian kernel with standard deviation

$$\sigma = 1.0 \quad (19)$$

was used in this experiment.

4.3.2 Mean Filter

The mean filter replaces each pixel with the average value of the surrounding neighborhood. A 3×3 kernel was used.

This operation smooths local intensity variations and can reduce small adversarial changes distributed across neighboring pixels.

4.3.3 Median Filter

The median filter replaces each pixel with the median value of its local neighborhood. A 3×3 kernel was used.

Unlike the mean filter, the median filter is less influenced by extreme pixel values and can therefore remove localized perturbations while preserving some image structure.

4.4 Denoising Results

Before applying any defense, the targeted FGSM attack produced

$$3510 \quad (20)$$

successful attacks out of

$$8965 \quad (21)$$

eligible test samples. This corresponds to a baseline targeted attack success rate of

$$39.15\%. \quad (22)$$

The three denoising defenses reduced the attack success rate.

Table 2: Targeted FGSM attack success rate after denoising defenses.

Defense Method	Successful Attacks	Attack Success Rate (%)
No Defense	3510 / 8965	39.15
Gaussian Blur	2420 / 8965	26.99
Mean Filter	2160 / 8965	24.09
Median Filter	2243 / 8965	25.02

Figure 6 compares the attack success rates for the different denoising defenses.

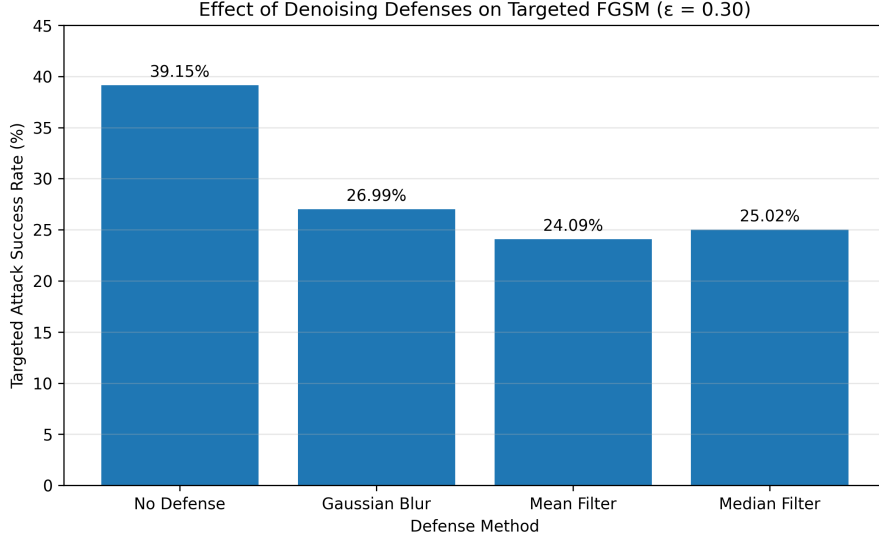


Figure 6: Targeted FGSM attack success rate after applying different denoising defenses at $\epsilon = 0.30$. Lower values indicate a more effective defense.

All three denoising methods reduced the effectiveness of the targeted FGSM attack.

Gaussian blur reduced the attack success rate from 39.15% to 26.99%. Median filtering reduced the success rate further to 25.02%.

The mean filter produced the best result among the tested denoising methods, reducing the targeted attack success rate to 24.09%.

Compared with the undefended baseline, this corresponds to a reduction of

$$39.15\% - 24.09\% = 15.06 \quad (23)$$

percentage points.

4.5 Gaussian Noise Defense

The second defense strategy consisted of adding random Gaussian noise to the adversarial images.

For an adversarial image x_{adv} , the defended image was generated as

$$x_{def} = x_{adv} + n, \quad (24)$$

where

$$n \sim \mathcal{N}(0, \sigma^2). \quad (25)$$

The parameter σ controls the magnitude of the added noise.

The following noise levels were evaluated:

$$\sigma = \{0.00, 0.02, 0.05, 0.10, 0.15, 0.20\}. \quad (26)$$

The purpose of this experiment was to determine whether random noise could disrupt the carefully selected gradient-based perturbation generated by FGSM.

4.6 Gaussian Noise Results

Table 3 shows the measured targeted attack success rate for each Gaussian noise level.

Table 3: Targeted FGSM attack success rate after adding Gaussian noise.

Noise Std. Dev. σ	Successful Attacks	Attack Success Rate (%)
0.00	3510 / 8965	39.15
0.02	3473 / 8965	38.74
0.05	3407 / 8965	38.00
0.10	3238 / 8965	36.12
0.15	3049 / 8965	34.01
0.20	2861 / 8965	31.91

Figure 7 shows the relationship between Gaussian noise level and targeted attack success rate.

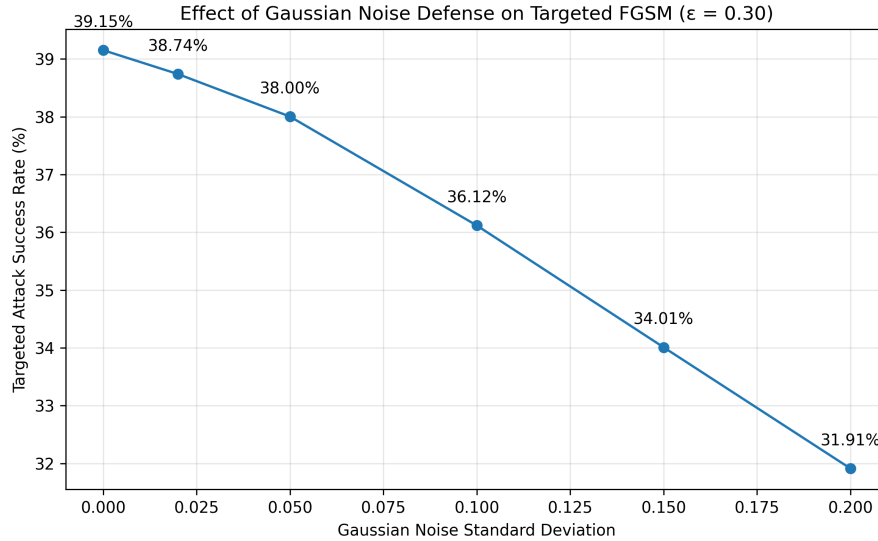


Figure 7: Effect of Gaussian noise magnitude on targeted FGSM attack success rate at $\epsilon = 0.30$.

The results show that increasing the Gaussian noise level consistently reduced the effectiveness of the targeted FGSM attack.

With no additional noise, the targeted attack success rate was 39.15%. A small noise level of $\sigma = 0.02$ produced only a small reduction, decreasing the attack success rate to 38.74%.

As the noise magnitude increased, the attack became progressively less effective. At $\sigma = 0.10$, the attack success rate decreased to 36.12%, while the largest tested noise level, $\sigma = 0.20$, reduced the attack success rate to 31.91%.

Although the Gaussian-noise defense reduced the attack success rate, it was less effective than the denoising filters. In particular, the mean filter reduced the attack success rate to 24.09%, compared with 31.91% for Gaussian noise with $\sigma = 0.20$.

4.7 Visual Comparison of Defense Methods

A successful targeted FGSM example was selected to visualize how the different defenses affected both the image and the neural network prediction.

Figure 8 shows the original image, the successful adversarial example, and the same adversarial image after each defense method was applied.

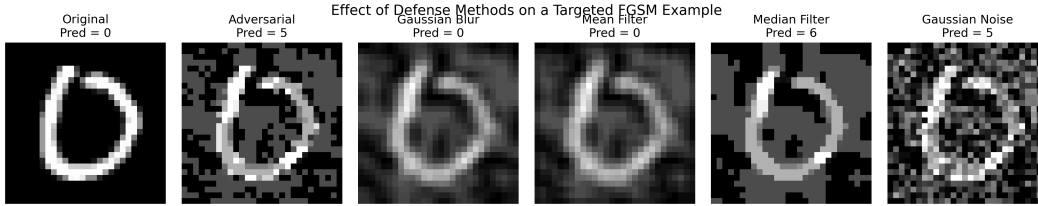


Figure 8: Visual comparison of defense methods applied to a successful targeted FGSM example. The original digit was classified as class 0, while the adversarial example was classified as target class 5.

The original sample was correctly classified as digit 0. After the targeted FGSM perturbation was applied, the classifier predicted the target digit 5.

Gaussian blur successfully removed enough of the adversarial perturbation for the classifier to recover the original prediction of 0. The mean filter also restored the prediction to the correct class 0.

The median filter prevented the model from predicting the target class 5, but the resulting prediction was class 6 rather than the original class 0. Therefore, the targeted attack was defeated, but the original classification was not recovered.

In contrast, the Gaussian-noise defense with $\sigma = 0.20$ did not defeat the targeted attack for this particular example, and the final prediction remained class 5.

This example illustrates that preventing a targeted adversarial attack does not necessarily imply that the original classification will be restored.

4.8 Discussion

The results demonstrate that simple preprocessing defenses can reduce the effectiveness of a targeted FGSM attack.

All three denoising techniques produced a substantial reduction in the attack success rate. Among the tested methods, mean filtering achieved the lowest attack success rate of 24.09%, followed by median filtering at 25.02% and Gaussian blur at 26.99%.

Adding Gaussian noise also reduced the targeted attack success rate, but the improvement was smaller. Increasing the noise standard deviation from 0 to 0.20 reduced the attack success rate from 39.15% to 31.91%.

These results suggest that smoothing-based preprocessing was more effective than random-noise injection for the tested targeted FGSM attack.

However, defense effectiveness must also be considered together with image quality. Strong preprocessing may remove adversarial perturbations, but it can also distort useful features in the original image. The visual comparison demonstrates this tradeoff. Gaussian blur and mean filtering successfully restored the correct prediction for the selected sample, while median filtering defeated the targeted attack but changed the prediction to an incorrect non-target class.

Overall, the experiment demonstrates that preprocessing can reduce the success of gradient-based adversarial attacks, but none of the tested defenses completely eliminated the attack. The results also highlight the importance of evaluating both attack success rate and classification quality when comparing adversarial defenses.

5 Overall Discussion

The three experiments demonstrate how adversarial perturbations can affect a neural-network classifier and how simple preprocessing techniques can reduce, but not completely eliminate, their impact.

In Task 1, the standard untargeted FGSM attack showed that increasing the perturbation magnitude ϵ reduced the classification performance of the MNIST model. As the perturbation became stronger, more images crossed a decision boundary and were classified incorrectly. This demonstrated that relatively small, gradient-based changes to the input can significantly affect the output of a neural network.

Task 2 extended the experiment from an untargeted attack to a targeted attack. Instead of simply forcing the classifier away from the correct class, the perturbation was generated specifically to move predictions toward class 5. This is a more restrictive objective because the attack must move an input toward one specific class rather than toward any incorrect class.

The targeted FGSM results showed a strong relationship between perturbation magnitude and attack success rate. At $\epsilon = 0$, the targeted attack success rate was 0%, as expected. The success rate increased to 2.05% at $\epsilon = 0.10$, 14.03% at $\epsilon = 0.20$, and 39.15% at $\epsilon = 0.30$.

These results indicate that larger perturbations make it easier to move an input toward the desired target class. However, the attack did not reach a 100% success rate even at the highest tested value of ϵ . This reflects the greater difficulty of a targeted attack compared with an untargeted attack.

Task 3 investigated whether simple preprocessing defenses could reduce the effectiveness of the targeted attack. Without any defense, the attack success rate at $\epsilon = 0.30$ was 39.15%. All three denoising approaches reduced this value.

The mean filter produced the best result, reducing the attack success rate to 24.09%. Median filtering produced a similar result of 25.02%, while Gaussian blur reduced the success rate to 26.99%.

Gaussian-noise injection also reduced the targeted attack success rate, but the effect was smaller. Increasing the noise standard deviation from $\sigma = 0$ to $\sigma = 0.20$ reduced the success rate from 39.15% to 31.91%.

A comparison of the tested defenses is summarized in Table 4.

Table 4: Comparison of the strongest tested defense configurations against targeted FGSM at $\epsilon = 0.30$.

Method	Targeted Attack Success Rate (%)
No Defense	39.15
Gaussian Blur	26.99
Mean Filter	24.09
Median Filter	25.02
Gaussian Noise ($\sigma = 0.20$)	31.91

Among the tested methods, the mean filter provided the largest reduction in targeted attack success. Compared with the undefended baseline, the attack success rate decreased by 15.06 percentage points.

The image examples also demonstrated that a defense can have several possible outcomes. In some cases, preprocessing restored the original classification. In other cases, it prevented the targeted prediction of class 5 but caused the network to predict a different incorrect class. Therefore, reducing targeted attack success does not necessarily mean that the classifier has fully recovered the correct prediction.

Another important observation is the tradeoff between defense strength and image quality. Denoising and noise injection can reduce adversarial perturbations, but they also modify the underlying input. If

the preprocessing becomes too strong, useful features of the original image may also be removed or distorted.

Overall, these experiments demonstrate that adversarial robustness cannot be evaluated only by measuring whether an attack succeeds. The effect of the defense on normal classification behavior and input quality must also be considered.

6 Conclusion

This project investigated adversarial example generation using the Fast Gradient Sign Method and evaluated several simple defenses against the attack.

First, an untargeted FGSM attack was applied to a pretrained MNIST classifier. The experiment demonstrated that increasing the perturbation magnitude reduced the classifier’s ability to correctly identify handwritten digits.

The attack was then modified into a targeted FGSM attack with the goal of forcing inputs toward class 5. The targeted attack success rate increased as the perturbation magnitude increased, reaching 39.15% at $\epsilon = 0.30$.

Finally, several preprocessing defenses were evaluated against the targeted attack. Gaussian blur, mean filtering, median filtering, and Gaussian-noise injection all reduced the attack success rate. The mean filter was the most effective tested defense, reducing the targeted attack success rate from 39.15% to 24.09%.

The results show that relatively simple preprocessing methods can reduce the effectiveness of adversarial attacks. However, none of the tested defenses completely prevented the targeted FGSM attack. In addition, stronger preprocessing can distort the input image and may affect normal classification performance.

The experiments therefore demonstrate both the vulnerability of neural networks to gradient-based adversarial perturbations and the difficulty of designing defenses that improve robustness without damaging useful input information.

References

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